

Alex Jacobs

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Education

University of Waterloo

Candidate for Bachelor of Applied Science in Mechatronics Engineering
Degree focus: Robotics and Automation, cGPA: 4.00/4.00

Sep. 2024 – Expected Jul. 2029
Waterloo, ON

Experience

Mechanical Designer – Quadrupedal Robotics

Sep. 2025 – Dec. 2025

Field AI

Irvine, CA

- Led end-to-end development of a modular, back-mounted backpack payload for the Unitree G1 humanoid robot, allowing for rapid testing and development of software and hardware components.
- Developed and tested ingress protection solutions, increasing the robot's viable environments.
- Designed and manufactured a modular, configurable, and robust sensor testing tower using rapid prototyping techniques, allowing for rapid testing and development of new sensor components.
- Developed testing frameworks and apparatus for validating sensors, allowing for optimal component selection.

Research Assistant – Mobile Robotics & SLAM

Jan. 2025 – Apr. 2025

Sensing and Robotics for Infrastructure Lab, University of California, Los Angeles

Los Angeles, CA

- Developed a custom Gazebo simulator in C++ and Python to collect LiDAR, visual, and inertial sensor data from a quadrupedal robot performing waypoint navigation.
- Developed a metric evaluation framework in Python to benchmark state-of-the-art SLAM on open datasets.
- Debugged and tested proprietary laboratory software for robotic sensor data collection, processing, and visualization.
- Designed and 3D printed SolidWorks models to integrate new modules into ground and aquatic robot platforms.

Projects

Back-Mounted Sensing and Compute Payload

Nov. 2025 – Dec. 2025

Field AI

Irvine, CA

- Owned end-to-end mechanical design, wiring, and thermal integration of a rigid backpack payload for the Unitree G1 humanoid, allowing existing payload hardware and software to be used on a new robotic platform.
- Rapidly designed and fabricated multiple backpack-style payloads via 3D printing to support parallel cross-office testing under tight development timelines.
- Produced comprehensive design documentation and file structure guides, enabling efficient handoff to downstream engineers.

Sensor Tower Test Rig

Oct. 2025 - Dec. 2025

Field AI

Irvine, CA

- Designed and manufactured a modular rigid sensor tower supporting LiDAR, camera, ultrasonic, and compute payload testing with configurable extrinsics.
- Built custom 3D printed fixtures and a Teensy 4.0-based validation framework to streamline sensor validation and characterization.
- Applied rapid prototyping methods to deliver a robust and configurable test platform for perception development.

Unmanned Surface Vehicle

Aug. 2023, Jan. 2025 – Apr. 2025

Sensing and Robotics for Infrastructure Lab, University of California, Los Angeles

Los Angeles, CA

- Maintained and extended SolidWorks models of the platform's above- and below-water sensing kits.
- Modeled and 3D printed a hydrodynamic fairing to reduce drag around the vehicle's underwater sensors.
- Assisted with data collection using detachable above-water sensors for a visual-inertial navigation ablation study.

Technical Skills

CAD Programs: SolidWorks, Onshape, Fusion 360, Blender | **Programming Languages:** C++, Python | **Manufacturing:** 3D printing, Lathe, Milling Machine, Drill Press, Band Saw, Scroll Saw

Awards & Achievements

Term Distinction, University of Waterloo, Waterloo, ON.

Fall 2024, Summer 2025

President's Scholarship of Distinction, University of Waterloo, Waterloo, ON.

May 2024